🎓 Assignment: No-Code Big Data Analytics for Science Communication

Total Weight: 40%

- **Part 1 (20%) – Wireframe & User Journey**
- **Part 2 (20%) – Analytics Interpretation & Value Proposition**
- 🟧 **Bonus (Up to 5%) – Prototype Implementation** *(Only for Part 1)*

🎯 Objective

Students will propose a web or mobile application that leverages **Big Data analytics** for a **Science Communication** use case.

They will design the **structure** (via wireframes), define the **user journey**, and explain how the application uses Big Data to benefit users.

No coding is required — students may use any **no-code** or **design platforms**.

📦 Part 1 – Proposal, Wireframe & User Journey (20%)

Deliverable: A presentation showcasing the proposed app concept, its use of big data, wireframes, user journey, and supported by slides to explain the overall idea clearly.

🔹 What You Need to Do:

1. **Define the Use Case & Problem Statement**
    - Choose a Science Communication scenario where **Big Data analytics utilized.**
    - Examples:
        - Analyzing public engagement trends in science-related news.
        - Identifying misinformation patterns in scientific topics.

        - Understanding how different audiences react to climate change reports.
        - Analyzing public interest trends in scientific topics

2. **Design the App/Web Concept**
    - Propose an app or web platform that implements the use case defined in (1).
    - You may describe it through:
        - High-level concept
        - Key features
        - Target users and pain points
    - Explain the data source(s), how data will be used, and the benefit to users
    - Example concept:

> ***Concept:*** *A personalized science news app that recommends articles based on real-time public interest trends.*
>
> ***How it works:***
>
> - *Tracks trending scientific keywords from platforms like Twitter, Google Trends, and Reddit (e.g., "fusion energy," "AI in medicine")*
> - *Analyzes which science topics are getting attention in different regions or age groups*
> - *Recommends curated news articles, explainer videos, or verified sources based on those trends*
>
> ***Big Data Use:***
>
> - *Aggregates large-scale user interest data*
> - *Matches user profiles with rising science topics*
> - *Learns from user reading behavior to refine future recommendations*

3. **Create a Wireframe**
    - Sketch or design the layout of your app/web:
        - Home screen
        - Interaction areas (upload, explore, filters, etc.)
        - Any key views (e.g., user dashboard, data results page)
    - Use tools like **Figma**, **Miro** or **Canva**
    - Data visualization or dashboard are *optional* — not all apps need dashboards (as long as data use is justified)

4. **Map the User Journey**
    - Show how a user interacts with the app/web, step by step:
        - Entry point (e.g., login, landing page)
        - Key actions (e.g., select topic, access reports)
        - End result (e.g., gained insights, downloaded summary)
5. **Prepare a Slide (Recommended)**
    - Create a simple slide to assist your presentation in explaining the app idea clearly
    - The slides should help illustrate:
        - The problem you're solving
        - Your proposed solution
        - How big data is used in the app
        - Key wireframe and user journey highlights
    - You may use any tool such as **Google Slides**, **PowerPoint** or **Gamma**

Note: The slides are not graded separately but are strongly recommended to support clarity during presentation.

🟧 BONUS (Up to 5%) – Prototype (Only if Part 1 is <20%)

- Students who **do not score full marks** on Part 1 can earn up to **+5% bonus** by building a **basic functional prototype**.
- Use **any tools**, including:
    - 🧱 **No-code builders**: Bubble.io, GlideApps, Retool, Graphite Note
    - 💻 **Basic coded front-ends** (if preferred)
- The prototype can be a **clickable mockup** or an **early-stage working interface**

📝 *Note: This bonus is only awarded if your Part 1 mark is below 20%. It cannot push your score above 20%.*

📦 Part 2: Analytics Interpretation & Value Proposition (20%)

Deliverable: A presentation explaining the data insights, visualizations, and the value of the application.

🔸 *Note: The dataset used in this part does not need to match the use case proposed in Part 1. You may explore a different dataset and topic, as long as it is relevant to Science Communication.*

🔹 What Students Need to Do:

1. **Choose a Dataset**
   - Find a real-world dataset related to Science Communication
   - Suggested sources:
     - Kaggle
     - Our World in Data
     - Data.gov
2. **Analyze the Data Using No-Code Tools**
   - Explore trends, correlations, and patterns
   - Recommended tools:
     - Google Looker Studio
     - Flourish Studio
     - Power BI
3. **Interpret the Results**
   - What insights can be drawn from the data?
   - How do these insights contribute to Science Communication?
   - Example interpretations:
     - "Videos generate 50% more engagement than text-based articles."
     - "Climate concern spikes after major disasters based on engagement data."
4. **Present the Value Proposition**
   - Why does this data matter?
   - Who benefits from these insights? (e.g., scientists, journalists, policymakers)
   - How can these findings be applied? (e.g. better science outreach strategies)

🧰 Suggested Tools by Category

Purpose	Recommended Tools
Wireframing	Figma, Miro, Canva
User Journey Mapping	*Same as your wireframing tool*
Analytics & Dashboards	Google Looker Studio, Flourish Studio, Power BI
Bonus: Prototyping	Bubble.io, GlideApps, Retool, Graphite Note

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🎤 Presentation Requirements

1. **Each group will be given 5 minutes** to present their work for each part of the assignment (Part 1 and Part 2).
2. **All group members must present at least once** across either Part 1 or Part 2.
3. **Prepare all necessary materials** to support your presentation. Refer to the assignment guidelines for what should be included.
4. **Ensure your files and links are ready and accessible before your presentation:**
    - Upload all materials to a **shared cloud folder** (e.g., Google Drive or OneDrive).
    - Test all links in advance to avoid technical delays.
5. **Be prepared for a brief Q&A session** after your presentation. The instructor may ask questions to clarify your concept, data interpretation, or implementation choices.
6. **Presentations will be evaluated not only on content but also on the quality of delivery.** This includes:
    - Clarity and confidence in speaking
    - Visual support and storytelling
    - Logical flow and time management

👉 *Refer to the rubric for full evaluation criteria.*